

120 Years of Olympic History: Participation and Medal Trends

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Introduction

Olympic Games:

- ▶ The most popular international sports competition every four years
 - ▶ Over 200 nations participate, with over 10,000 athletes, and over 300 events
 - ▶ First modern Olympic Games were held in Athens, Greece in 1896
 - ▶ Have expanded enormously in scale, diversity and global importance
- Studying historical trends in Olympic participation and performance can shed light on wider trends in gender equality, national development and sporting investment. → Provide a data-based view on the evolution of the Olympic movement

Dataset + Research Questions

Dataset:

- ▶ Collected from Kaggle
- ▶ 271,000 observations of athletes, events, medals, and attributes (Summer and Winter Olympics)
- ▶ From 1896 to 2016 (120 years of Olympic history)

Research Questions:

- ▶ How has overall athlete participation in the Summer Olympic Games changed over time?
- ▶ How has female representation among Olympic athletes changed over time?
- ▶ Has the proportion of Olympic medals won by female athletes increased over time?
- ▶ Has the distribution of Olympic medals among nations become more or less concentrated over time?

Methodology

Visualisation of trends in participation and medals over time

Implementation of a simple linear regression model and logistic regression model to test specific hypotheses about the relationship

1. Overall participation patterns over time:

▶ $Participation_t \sim \beta_0 + \beta_1 * Year + \epsilon_t$

2. Female Representation over time:

▶ $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 * Year + \epsilon_t$

3. Female Medal Success over time:

▶ $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 * Year + \epsilon_t$

4. Medal Concentration Among Nations:

▶ $HHI_t \sim \beta_0 + \beta_1 * Year + \epsilon_t$

Results

Visualisation

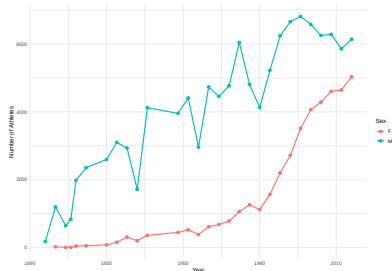


Figure 1: Athlete Participation by Gender in the Summer Olympics (1896–2016)

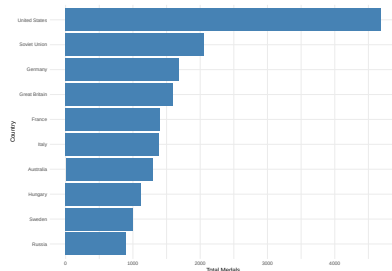


Figure 2: Top 10 Countries by Total Medal Count in the Summer Olympics (1896–2016)

Visualisation

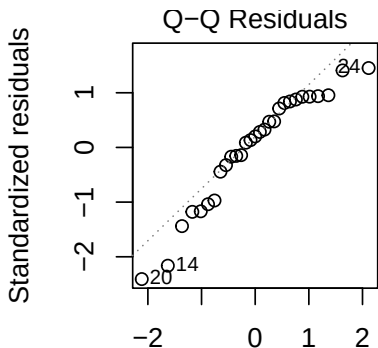
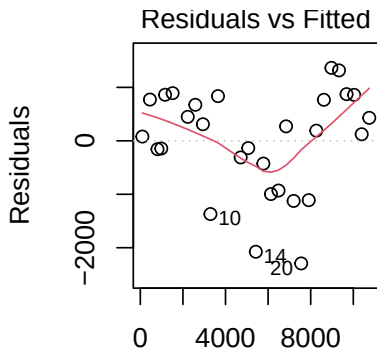
Table 1: Top 5 Countries by Total Medals (Summer Olympics)

Country	Total Medals
United States	4686
Soviet Union	2061
Germany	1687
Great Britain	1598
France	1408

Overall participation patterns over time

Table 2

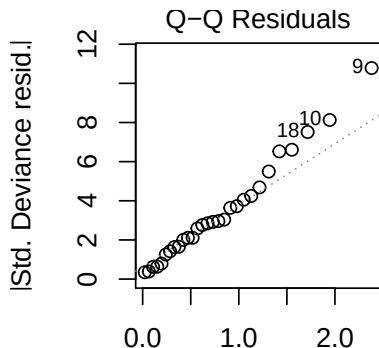
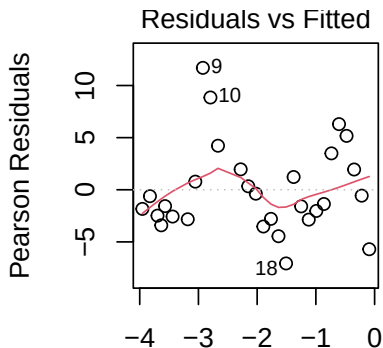
Term	Coefficient	p-value	R ²
Intercept	96.6205	0.7825	0.924784415470121
Year	88.7696	0.0000	



Female Representation over time

Table 3

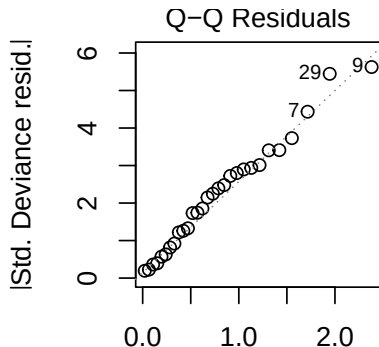
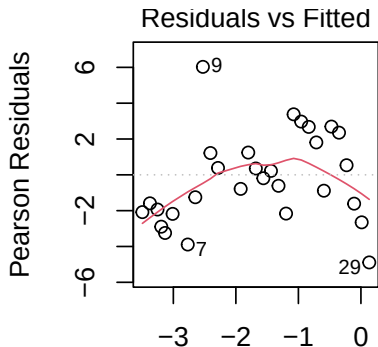
Term	Coefficient	p-value	Pseudo R ²
Intercept	-3.9548	0	0.974581359270954
Year	0.0322	0	



Female Medal Success over time

Table 4

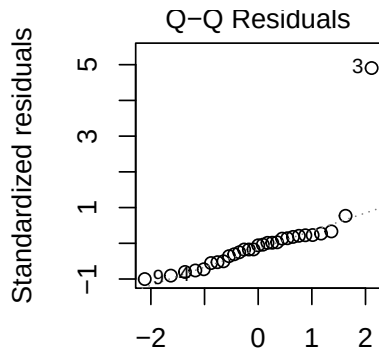
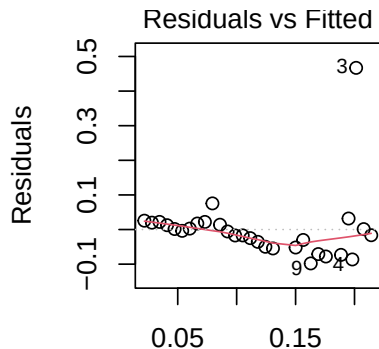
Term	Coefficient	p-value	Pseudo R ²
Intercept	-3.4954	0	0.964539393668614
Year	0.0302	0	



Medal Concentration Among Nations

Table 5

Term	Coefficient	p-value	R ²
Intercept	0.2142	0.0000	0.274710131339156
Year	-0.0016	0.0035	



Conclusion and Recommendation

- ▶ Athlete participation in the Summer Olympics has increased significantly over the past 120 years
- ▶ Female representation among Olympic athletes has increased dramatically, and the proportion of medals won by female athletes has also increased significantly over time
- ▶ The distribution of Olympic medals across nations has become less concentrated over time

Limitation: Not account for other factors and the linear and logistic regression models not fully capture non-linear changes

Future research: : Explore more complex models and conduct qualitative analyses to understand the social and political factors driving changes in Olympic participation and success